

Machine Learning Handbook

From Beginner to Interview Ready (Banglish Edition)

AI/ML Study Guide

2026

Contents

1 Chapter 1: Introduction to AI	5
1.1 1.1 Ki Ei AI, ML, DL? (Zero theke shuru)	5
1.2 1.2 AI vs ML vs DL — Comparison Table	6
1.3 1.3 Real World Examples	6
1.4 Summary	6
1.5 Common Mistakes	6
1.6 Interview Questions	7
1.7 Quick Revision	7
2 Chapter 2: Machine Learning Workflow	8
2.1 2.1 Data Collection	8
2.2 2.2 Data Cleaning	8
2.3 2.3 EDA (Exploratory Data Analysis)	8
2.4 2.4 Feature Engineering	8
2.5 2.5 Feature Selection	9
2.6 2.6 Train/Test Split	9
2.7 2.7 Model Training	9
2.8 2.8 Evaluation	9
2.9 2.9 Hyperparameter Tuning	9
2.10 2.10 Deployment	9
2.11 Summary	9
2.12 Common Mistakes	9
2.13 Interview Questions	10
2.14 Quick Revision	10
3 Chapter 3: Supervised Learning	11
3.1 3.1 Supervised Learning Ki?	11
3.2 3.2 Regression Ki?	11
3.3 3.3 Classification Ki?	11
3.4 3.4 Regression vs Classification — Comparison Table	11
3.5 Summary	11
3.6 Common Mistakes	12
3.7 Interview Questions	12
3.8 Quick Revision	12
4 Chapter 4: Regression Algorithms	13
4.1 4.1 Linear Regression	13
4.2 4.2 Polynomial Regression	14
4.3 4.3 Ridge Regression (L2 Regularization)	14

4.4	4.4 Lasso Regression (L1 Regularization)	15
4.5	4.5 ElasticNet Regression	15
4.6	4.6 Decision Tree Regressor	16
4.7	4.7 Random Forest Regressor	16
4.8	Summary	17
4.9	Common Mistakes (Overall Chapter)	17
4.10	Interview Questions (Overall)	17
4.11	Quick Revision	17
5	Chapter 5: Regression Metrics	18
5.1	5.1 MAE (Mean Absolute Error)	18
5.2	5.2 MSE (Mean Squared Error)	18
5.3	5.3 RMSE (Root Mean Squared Error)	18
5.4	5.4 R^2 Score (Coefficient of Determination)	18
5.5	5.5 Adjusted R^2 Score	19
5.6	Comparison Table	19
5.7	Summary	19
5.8	Common Mistakes	19
5.9	Interview Questions	19
5.10	Quick Revision	20
6	Chapter 6: Classification	21
6.1	6.1 Logistic Regression	21
6.2	6.2 Decision Tree (Classification)	21
6.3	6.3 Random Forest (Classification)	22
6.4	6.4 K-Nearest Neighbors (KNN)	23
6.5	6.5 Naive Bayes	23
6.6	6.6 Support Vector Machine (SVM)	24
6.7	Summary Table	24
6.8	Common Mistakes (Overall)	25
6.9	Interview Questions (Overall)	25
6.10	Quick Revision	25
7	Chapter 7: Classification Metrics	26
7.1	7.1 Confusion Matrix	26
7.2	7.2 Accuracy	26
7.3	7.3 Precision	26
7.4	7.4 Recall (Sensitivity)	26
7.5	7.5 Specificity	26
7.6	7.6 F1 Score	27
7.7	7.7 Confusion Matrix Calculation Example	27
7.8	7.8 ROC Curve & AUC	27
7.9	7.9 Precision-Recall Curve	27
7.10	Comparison Table	28
7.11	Summary	28
7.12	Common Mistakes	28
7.13	Interview Questions	28
7.14	Quick Revision	28
8	Chapter 8: Ensemble Learning	29
8.1	8.1 Ensemble Learning Ki?	29
8.2	8.2 Bagging (Bootstrap Aggregating)	29
8.3	8.3 Boosting	29
8.4	8.4 Random Forest (Recap)	29
8.5	8.5 AdaBoost (Adaptive Boosting)	29

8.6	8.6 Gradient Boosting	29
8.7	8.7 XGBoost (Extreme Gradient Boosting)	30
8.8	8.8 LightGBM	30
8.9	8.9 CatBoost	30
8.10	8.10 Comparison Table	30
8.11	Summary	31
8.12	Common Mistakes	31
8.13	Interview Questions	31
8.14	Quick Revision	31
9	Chapter 9: Hyperparameter Tuning	32
9.1	9.1 GridSearchCV	32
9.2	9.2 RandomizedSearchCV	32
9.3	9.3 Bayesian Optimization	32
9.4	9.4 Early Stopping	32
9.5	Comparison Table	32
9.6	Summary	33
9.7	Common Mistakes	33
9.8	Interview Questions	33
9.9	Quick Revision	33
10	Chapter 10: Cross Validation	34
10.1	10.1 Hold-Out Validation	34
10.2	10.2 K-Fold Cross Validation	34
10.3	10.3 Stratified K-Fold	34
10.4	10.4 Leave-One-Out (LOOCV)	34
10.5	Comparison Table	34
10.6	Summary	34
10.7	Common Mistakes	35
10.8	Interview Questions	35
10.9	Quick Revision	35
11	Chapter 11: Feature Engineering	36
11.1	11.1 Missing Values	36
11.2	11.2 Outlier Handling	36
11.3	11.3 Feature Scaling	36
11.4	11.4 Label Encoding	36
11.5	11.5 One-Hot Encoding	36
11.6	11.6 Target Encoding	37
11.7	11.7 Feature Selection	37
11.8	Comparison Table	37
11.9	Summary	37
11.10	Common Mistakes	37
11.11	Interview Questions	37
11.12	Quick Revision	38
12	Chapter 12: Complete Machine Learning Pipeline	39
12.1	12.1 Data Loading	39
12.2	12.2 EDA	39
12.3	12.3 Data Cleaning	39
12.4	12.4 Feature Engineering	39
12.5	12.5 Feature Selection & Split	39
12.6	12.6 Scaling	40
12.7	12.7 Model Building (Multiple Model Comparison)	40
12.8	12.8 Hyperparameter Tuning (Best Model)	40

12.9	12.9 Final Evaluation	41
12.10	12.10 Pipeline Flowchart (Full Recap)	41
12.11	12.11 Summary	41
12.12	12.12 Common Mistakes	41
12.13	12.13 Interview Questions	41
12.14	12.14 Quick Revision	41
13	Chapter 13: Interview Questions (100+ with Answers)	42
13.1A	13.1A. AI/ML Basics (Q1-10)	42
13.2B	13.2B. Data Preprocessing (Q11-25)	42
13.3C	13.3C. Regression (Q26-40)	43
13.4D	13.4D. Classification (Q41-60)	44
13.5E	13.5E. Metrics (Q61-70)	45
13.6F	13.6F. Ensemble & Boosting (Q71-80)	45
13.7G	13.7G. Model Selection & Tuning (Q81-90)	46
13.8H	13.8H. General / Advanced (Q91-100+)	46
14	Chapter 14: Cheat Sheet	48
14.1	14.1 One-Page ML Workflow	48
14.2	14.2 All Important Formulas	48
14.3	14.3 Algorithm Comparison (Regression)	48
14.4	14.4 Algorithm Comparison (Classification)	48
14.5	14.5 Metric Comparison	49
14.6	14.6 Bagging vs Boosting	49
14.7	14.7 Flowchart: Which Algorithm to Choose?	49
14.8	14.8 Common Interview Tricks	49
14.9	14.9 Final Quick Revision (Whole Book)	50

1 Chapter 1: Introduction to AI

1.1 1.1 Ki Ei AI, ML, DL? (Zero theke shuru)

Dhori, tumi ekta chotto bachcha ke hazar ta cat ar dog er chobi dekhaila. Kisudin por bachcha ta nije bolte parbe “eta cat” naki “eta dog” — kono rule na shikheo, sudhu pattern dekhe dekhe shikhse. **Machine Learning** thik eirokom — machine ke data dekhiye “shikhte” deoya hoy, hard-coded rule likhe deoya lagena.

Highlight Box AI ekta broad concept — “machine ke intelligent banano”. ML holo AI achieve korar ekta **method**. DL holo ML er ekta **subset**, jekhane deep neural network use hoy.

1.1.1 Artificial Intelligence (AI)

AI mane emon system toiri kora jeta manusher moto “intelligent” behavior dekhate pare — reasoning, decision making, problem solving.

- **Example:** Chess khela computer, Google Maps er shortest path, Siri/Alexa.
- AI-r vitore duita boro category ache: **Rule-based AI** (if-else diye likha, jemon old expert systems) ar **Learning-based AI** (data theke shikhe, eta-i ML).

1.1.2 Machine Learning (ML)

ML holo emon technique jekhane algorithm data theke pattern shikhe, explicitly program na kore prediction/decision dey.

- **Real example:** Netflix tumi ki dekho seta dekhe next recommendation dey — kew “if user watched horror then recommend horror” emon code likhe nai, model nijei pattern ber korse.

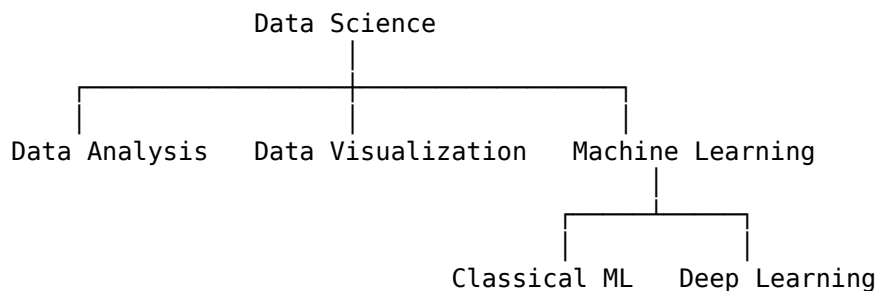
1.1.3 Deep Learning (DL)

DL holo ML-er ekta subset jekhane multiple layer wala **neural network** use hoy, jeta onek complex pattern (image, speech, text) shikhte pare.

- **Example:** Face unlock, ChatGPT/Claude er moto language model, self-driving car er object detection.

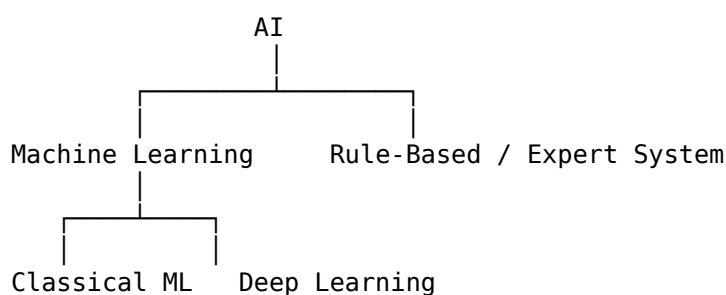
1.1.4 Data Science (DS)

Data Science holo aro boro ekta umbrella term — data collect kora, clean kora, analyze kora, visualize kora, ar tar upor ML model boshano — shob mile Data Science.



1.2 AI vs ML vs DL — Comparison Table

Feature	AI	ML	DL
Scope	Sobcheye boro concept	AI er subset	ML er subset
Goal	Intelligent behavior toiri kora	Data theke shikhe prediction	Complex pattern (image/speech/text) shikha
Data lagbe kina	Lagte pare, nao lagte pare	Lagbei	Onek boro dataset lagbe
Example	Chess AI, Rule-based chatbot	Spam detection, Price prediction	Face recognition, ChatGPT
Hardware	Normal	Normal	Powerful (GPU lagte pare)



1.3 Real World Examples

Application	AI/ML/DL Type	Explanation
Gmail Spam Filter	ML (Classification)	Email spam naki normal predict kore
Netflix Recommendation	ML	User behavior theke recommend kore
Face Unlock	DL (CNN)	Face image theke pattern shikhe
ChatGPT / Claude	DL (Transformer)	Language pattern shikhe text generate kore
Fraud Detection (Bank)	ML	Transaction pattern dekhe anomaly detect kore
Self-driving Car	DL	Camera/sensor data theke object detect kore

1.4 Summary

- AI holo broad goal — machine ke intelligent banano.
- ML holo data theke shikhe prediction deoya-r technique.
- DL holo neural network based ML, complex pattern-er jonno.
- Data Science holo puro pipeline — data theke insight/model porjonto.

1.5 Common Mistakes

- AI, ML, DL ke synonym mone kora — bhul. $ML \subset AI$, $DL \subset ML$.

- “AI mane robot” emon vabha — AI onek jaygay invisible-o thake (recommendation system, spam filter).
- Data Science ke sudhu “coding” mone kora — asholey eta statistics + coding + business understanding-er combination.

1.6 Interview Questions

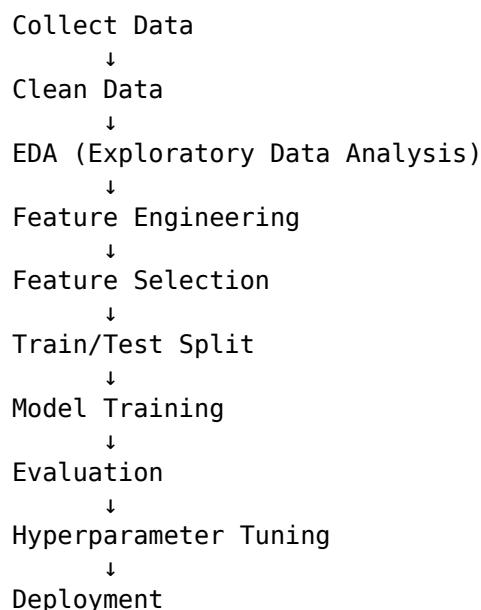
1. AI, ML, ar DL-er moddhe difference ki?
2. Ekta real-life example dao jekhane ML use hoy but DL na.
3. Deep Learning-er jonno keno boro dataset lage?
4. Rule-based system ar ML-based system-er moddhe difference ki?
5. Data Science ar Machine Learning ki same jinis?

1.7 Quick Revision

- AI > ML > DL (nesting order)
- ML = pattern shikha data theke
- DL = neural network diye complex pattern shikha
- DS = full pipeline (data → insight/model)

2 Chapter 2: Machine Learning Workflow

Kono ML project shuru theke deploy porjonto ekta fixed workflow follow kore. Eta na jene direct model banate gele shobshomoy fail hobe.



2.1 2.1 Data Collection

Data source hote pare — CSV file, database, API, web scraping, sensor. **Real example:** House price predict korte hole tomar lagbe location, area, bedroom, price emon historical data.

2.2 2.2 Data Cleaning

Real world data shobshomoy “messy” hoy — missing value, duplicate row, wrong datatype, inconsistent format thake.

- Missing value handle kora (drop / impute)
- Duplicate remove kora
- Wrong datatype fix kora (e.g. “25” string ke int banano)
- Outlier check kora

Tip: “Garbage in, Garbage out” — data jodi kharap hoy, model kokhono valo hobe na, algorithm jotoi advanced hok na keno.

2.3 2.3 EDA (Exploratory Data Analysis)

EDA mane data ke visually ar statistically bujha — distribution, correlation, pattern ber kora.

- `df.describe()`, `df.info()`, histogram, boxplot, correlation heatmap
- Target variable-er distribution dekha (balanced naki imbalanced)

2.4 2.4 Feature Engineering

Existing column theke notun useful column banano, jate model better shikhte pare.

- **Example:** “Date of Birth” theke “Age” feature banano.

- Domain knowledge use kora — jemon e-commerce e “Total Spend / Number of Orders = Avg Order Value” notun feature.

2.5 2.5 Feature Selection

Shob feature model-er jonno useful na — kichu feature noise create kore. Correlation, feature importance, statistical test diye best feature bacha hoy.

2.6 2.6 Train/Test Split

Model ke evaluate korar jonno data ke train ar test e bhag kora hoy — jate “unseen” data-r upor model kemon perform kore bujha jay.

```
from sklearn.model_selection import train_test_split
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)
```

Important Formula Box Typical split: 80% Train, 20% Test (or 70/30). Boro dataset hole 90/10 o kora jay.

2.7 2.7 Model Training

Algorithm choose kore train data diye model ke “fit” kora hoy — model data theke pattern/weight shikhe.

2.8 2.8 Evaluation

Test data diye model kemon perform korche seta measure kora — Regression e RMSE/R², Classification e Accuracy/F1 etc (Chapter 5, 7 e details).

2.9 2.9 Hyperparameter Tuning

Model-er best configuration khoja — jemon Random Forest e koyta tree hobe, Decision Tree koto depth porjonto jabe (Chapter 9 e details).

2.10 2.10 Deployment

Trained model ke real world application e boshano — API banano, web app e integrate kora, ba mobile app e use kora.

2.11 Summary

- ML workflow ekta cyclic, structured process — kono step skip korle model unreliable hoye jay.
- Data quality ar feature engineering-i ashole sobcheye boro impact fele, algorithm choice tar por.

2.12 Common Mistakes

- EDA skip kore shortcut e model training e chole jawa.
- Train/Test split age na kore, pura dataset diye feature engineering kora (**data leakage** hoy).

- Feature scaling na kore distance-based algorithm (KNN, SVM) use kora.

2.13 Interview Questions

1. ML workflow-er steps gula bolo order soho.
2. Data leakage ki, ki kore hoy?
3. EDA korar importance ki?
4. Train/Test split keno dorkar?
5. Feature Engineering ar Feature Selection-er moddhe difference ki?

2.14 Quick Revision

Collect → Clean → EDA → Feature Engineering → Selection → Split → Train → Evaluate → Tune
→ Deploy

3 Chapter 3: Supervised Learning

3.1 3.1 Supervised Learning Ki?

Supervised Learning mane amra model ke **input (X)** ar tar shathe correct **output/label (y)** dui-i dei — model eita theke input-output relationship shikhe.

Analogy: Eta thik teacher-er shathe student-er moto — teacher question dey answer soho, student pattern shikhe, porer bar notun question ashle nijei answer korte pare.

Supervised Learning-er duita main type ache: **Regression** ar **Classification**.

3.2 3.2 Regression Ki?

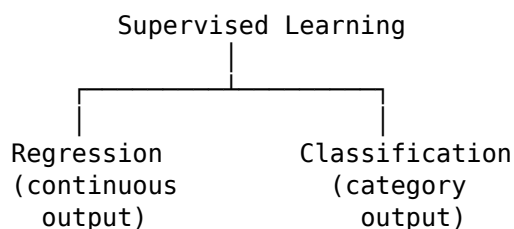
Jokhon output **continuous number** (jekono value hote pare), take Regression bola hoy.

- Example: House price prediction (500000, 750000...), Temperature prediction, Salary prediction

3.3 3.3 Classification Ki?

Jokhon output **discrete category** (fixed set-er moddhe ekta), take Classification bola hoy.

- Example: Email Spam/Not Spam, Disease Positive/Negative, Cat/Dog/Rabbit



3.4 3.4 Regression vs Classification — Comparison Table

Feature	Regression	Classification
Output type	Continuous number	Discrete category
Example	Price, Age, Salary prediction	Spam/Not Spam, Disease Yes/No
Algorithms	Linear Regression, Ridge, Lasso	Logistic Regression, KNN, SVM
Evaluation Metric	MAE, MSE, RMSE, R ²	Accuracy, Precision, Recall, F1
Output example	75000.50	“Spam” or 1
Decision boundary	Nai (line/curve fit kore)	Ache (categories alada kore)

Interview Note: Kew jodi jiggesh kore “Age prediction Regression na Classification?” — Age jodi exact number (23.5 years) hoy tahole Regression, kintu jodi “Age Group: Child/Adult/Senior” hoy tahole Classification.

3.5 Summary

- Supervised Learning e label soho data diye model train kora hoy.
- Regression → continuous output, Classification → category output.

3.6 Common Mistakes

- Regression problem ke Classification hisebe treat kora (ba ulta) — problem statement bhalo kore na bujha.
- Continuous target ke unnecessarily bin/category kore accuracy kome jawa.

3.7 Interview Questions

1. Supervised Learning ki, Unsupervised theke ki difference?
2. Regression ar Classification-er moddhe main difference ki?
3. Ekta problem dile bolte parbe eta Regression na Classification? (e.g. “predict number of items sold”)
4. Kono algorithm ki Regression ar Classification duitai korte pare? (Yes — Decision Tree, Random Forest, KNN)

3.8 Quick Revision

Supervised = Input + Label diye train → Regression (number) | Classification (category)

4 Chapter 4: Regression Algorithms

4.1 4.1 Linear Regression

Intuition: Ekta straight line fit kora hoy data points-er moddhe diye, jate line theke actual point-er distance (error) sobcheye kom hoy.

Real-life example: Ghar-er size (sqft) diye price predict kora — size barle price o mota-muti linearly barbe.

Mathematics: Amra emon ekta line khoji jeta best fit kore. Line-er equation:

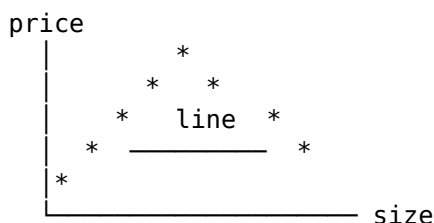
$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_n x_n + \epsilon$$

Ekhane β_0 = intercept, $\beta_1 \dots \beta_n$ = coefficients (weight), ϵ = error term.

Formula (Cost Function) — Mean Squared Error minimize kora hoy:

$$J(\beta) = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

Optimization: Gradient Descent ba Normal Equation diye best β ber kora hoy jate $J(\beta)$ minimum hoy.



Assumptions: - Linearity (X ar y-er relationship linear) - No multicollinearity (features nijeder moddhe highly correlated na hobe) - Homoscedasticity (error-er variance constant thakbe) - Residuals normally distributed

Advantages: Simple, fast, interpretable, kom data-teo kaj kore. **Disadvantages:** Non-linear relationship bujhte pare na, outlier-e sensitive.

Hyperparameters: fit_intercept, n_jobs (mostly no strong hyperparameter, tai regularized version-er dorkar hoy)

```
from sklearn.linear_model import LinearRegression
from sklearn.model_selection import train_test_split
from sklearn.metrics import mean_squared_error, r2_score
```

```
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)
```

```
model = LinearRegression()
model.fit(X_train, y_train)
```

```
y_pred = model.predict(X_test)
print("R2:", r2_score(y_test, y_pred))
print("MSE:", mean_squared_error(y_test, y_pred))
```

Interview Questions: OLS ki? Multicollinearity keno problem? Gradient Descent kivabe kaj kore? **Common Mistakes:** Feature scaling na kora (gradient descent slow hoy), outlier na check kora, categorical variable ke encode na kore direct use kora.

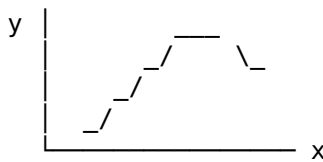
4.2 4.2 Polynomial Regression

Intuition: Data jodi curve shape-e thake (straight line fit na hoy), tahole feature-er higher power (x^2 , x^3 ...) add kore curve fit kora hoy.

Real-life example: Gari-r speed ar fuel efficiency — relationship curve-shaped hote pare.

Formula:

$$y = \beta_0 + \beta_1 x + \beta_2 x^2 + \dots + \beta_n x^n$$



Advantages: Non-linear pattern capture korte pare. **Disadvantages:** High degree diele **overfitting** hoy shohoje.

Hyperparameters: degree (kototuku power porjonto jabe)

```
from sklearn.preprocessing import PolynomialFeatures
from sklearn.linear_model import LinearRegression

poly = PolynomialFeatures(degree=3)
X_poly = poly.fit_transform(X_train)

model = LinearRegression()
model.fit(X_poly, y_train)
```

Interview Questions: Degree barale ki hoy? Overfitting kivabe control kora jay eikhane? **Common Mistakes:** Very high degree use kore overfit kora, test set-e evaluate na kora.

4.3 4.3 Ridge Regression (L2 Regularization)

Intuition: Linear Regression-e jodi overfitting hoy (onek feature, coefficient onek boro), Ridge shei coefficient-gulake “chotto” kore rakhe, jate model simple thake.

Formula:

$$J(\beta) = \sum_{i=1}^n (y_i - \hat{y}_i)^2 + \lambda \sum_{j=1}^p \beta_j^2$$

λ (alpha) hoche regularization strength — jotobeshi, totobeshi coefficient chotto hobe.

Advantages: Overfitting kome, multicollinearity handle kore. **Disadvantages:** Coefficient kokhono ekdome 0 hoy na — feature selection hoy na.

Hyperparameters: alpha (regularization strength)

```
from sklearn.linear_model import Ridge
model = Ridge(alpha=1.0)
model.fit(X_train, y_train)
```

Interview Questions: Ridge kivabe overfitting reduce kore? Alpha barale/komale ki hoy?
Common Mistakes: Alpha na tune kore default value use kora.

4.4 4.4 Lasso Regression (L1 Regularization)

Intuition: Ridge-er moto, kintu Lasso kichu coefficient ke **ekdom 0** kore dite pare — mane automatic feature selection hoye jay.

Formula:

$$J(\beta) = \sum_{i=1}^n (y_i - \hat{y}_i)^2 + \lambda \sum_{j=1}^p |\beta_j|$$

Advantages: Feature selection automatically hoy (sparse model). **Disadvantages:** Highly correlated feature thakle random-e ekta rekhe baki gula 0 kore dey — unstable hote pare.

Hyperparameters: alpha

```
from sklearn.linear_model import Lasso
model = Lasso(alpha=0.1)
model.fit(X_train, y_train)
```

Interview Questions: Ridge ar Lasso-r moddhe main difference ki? Kokhon Lasso use korba?
Common Mistakes: Lasso ke boro dataset-e high alpha diye shob feature 0 kore fela.

4.5 4.5 ElasticNet Regression

Intuition: Ridge (L2) ar Lasso (L1) duitar combination — duitar sub-difat pawa jay.

Formula:

$$J(\beta) = \sum (y_i - \hat{y}_i)^2 + \lambda_1 \sum |\beta_j| + \lambda_2 \sum \beta_j^2$$

Hyperparameters: alpha, l1_ratio (koto percent L1 vs L2)

```
from sklearn.linear_model import ElasticNet
model = ElasticNet(alpha=0.1, l1_ratio=0.5)
model.fit(X_train, y_train)
```

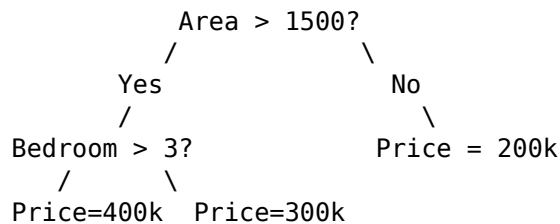
Advantages: Lasso-r instability kome, Ridge-er stability + Lasso-r feature selection duitai pawa jay. **Disadvantages:** Duita hyperparameter tune korte hoy — complexity barse.

Interview Questions: L1_ratio=1 hole eta ki hobe? (Answer: pure Lasso) L1_ratio=0 hole? (pure Ridge)

4.6 Decision Tree Regressor

Intuition: Data ke bar bar “question” diye split kora hoy (jemon “Area > 1000 sqft?”), protita leaf node e average value prediction hishebe deoya hoy.

Real-life example: Ghar-er price predict korte “location ki city center?”, “bedroom koyta?” emon question diye tree banano.



Splitting Criterion: Variance Reduction / MSE reduction — jei split-e sobcheye beshi variance kome, shei split e nea hoy.

Advantages: Non-linear relationship shikhte pare, interpretable, scaling lage na. **Disadvantages:** Overfit korar tendency onek — ekta chotto data change e tree onek change hoy.

Hyperparameters: max_depth, min_samples_split, min_samples_leaf

```
from sklearn.tree import DecisionTreeRegressor
model = DecisionTreeRegressor(max_depth=5, min_samples_split=10)
model.fit(X_train, y_train)
```

Interview Questions: Decision Tree-e overfitting kivabe rodh korba? Regression tree-e splitting criteria ki? (Variance/MSE reduction, Classification e Gini/Entropy) **Common Mistakes:** max_depth set na kore full tree grow korte deoya — heavy overfitting hoy.

4.7 Random Forest Regressor

Intuition: Onek gulo Decision Tree banano hoy (each tree different random data subset + random feature diye), tarpor shob tree-r prediction-er **average** neya hoy. Eta “wisdom of the crowd” concept.

Tree 1 → 210k
Tree 2 → 195k → Average → 205k (Final Prediction)
Tree 3 → 210k

Advantages: Single tree theke onek stable, overfitting kom, high accuracy. **Disadvantages:** Slow (onek tree), interpretability kome jay (“black box”-er kache).

Hyperparameters: n_estimators (koyta tree), max_depth, max_features

```
from sklearn.ensemble import RandomForestRegressor
model = RandomForestRegressor(n_estimators=200, max_depth=10, random_state=42)
model.fit(X_train, y_train)
```

Interview Questions: Random Forest-e “random” ki duita jaygay ase? (Bootstrap sampling of data + Random feature subset). Bagging ki? **Common Mistakes:** n_estimators khub kom rakha (unstable prediction), features scale korar dorkar nai eta bhule scale kore extra kaj kora.

4.8 Summary

Algorithm	Best jokhon	Regularization
Linear Regression	Simple linear relationship	Nai
Polynomial Regression	Curved relationship	Nai
Ridge	Multicollinearity ache	L2
Lasso	Feature selection lage	L1
ElasticNet	Onok feature + correlated	L1+L2
Decision Tree	Non-linear, interpretable lage	Nai (pruning diye)
Random Forest	High accuracy lage, interpretability kom important	Ensemble

4.9 Common Mistakes (Overall Chapter)

- Feature scaling na kora Linear/Ridge/Lasso-r jonno (distance/gradient based).
- Overfitting check na kore direct train accuracy dekhe khushi hoye jawa.
- Regularization-er alpha tune na kora.

4.10 Interview Questions (Overall)

1. Regularization ki, keno lage?
2. Bias-Variance tradeoff regression-er context-e explain koro.
3. Random Forest single Decision Tree theke better keno?
4. Ridge vs Lasso — kokhon kon ta use korba?

4.11 Quick Revision

Linear → simple line | Polynomial → curve | Ridge(L2)/Lasso(L1)/ElasticNet → regularized |
Decision Tree → rule based split | Random Forest → many trees average

5 Chapter 5: Regression Metrics

Model banano hoye gele, model kemon perform korche seta measure korte hoy — er jonno regression metrics use kora hoy.

5.1 MAE (Mean Absolute Error)

Formula:

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|$$

Intuition: Actual ar predicted value-r moddhe average “absolute” difference. **When to use:** Jokhon outlier-er effect kom rakhte chao. **Advantages:** Bujhte shohoj, outlier-e less sensitive. **Disadvantages:** Boro error-ke penalize kore na beshi (linear penalty). **Real-world example:** Average e prediction koto taka off hocche seta bujha — jemon “average 5000 taka error”.

5.2 MSE (Mean Squared Error)

Formula:

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

Intuition: Error ke square kore average kora hoy — boro error ke onek beshi penalize kore. **When to use:** Jokhon boro error ke beshi “kharap” hishebe treat korte chao. **Advantages:** Mathematically differentiable, optimization-e use kora shohoj. **Disadvantages:** Unit square hoye jay (jemon taka² — interpret korte kothin), outlier-e highly sensitive.

5.3 RMSE (Root Mean Squared Error)

Formula:

$$RMSE = \sqrt{MSE}$$

Intuition: MSE-r square root neya hoy jate unit abar original scale-e chole ashe (taka² theke taka). **When to use:** Sobcheye common metric — real-world interpretable ar boro error penalize korte pare. **Real example:** RMSE = 5000 taka mane bolte pare, average error 5000 taka-r kache-kachi (kintu MAE theke bit different weighting).

5.4 R² Score (Coefficient of Determination)

Formula:

$$R^2 = 1 - \frac{\sum (y_i - \hat{y}_i)^2}{\sum (y_i - \bar{y})^2}$$

Intuition: Model koto percent variance explain korte parche, mean-only baseline-er tulonay. R²=1 mane perfect fit, R²=0 mane model just mean predict korche, negative hole model baseline thea kharap. **When to use:** Model-er overall “goodness of fit” bujhte.

5.5 Adjusted R² Score

Formula:

$$Adjusted R^2 = 1 - \left[\frac{(1 - R^2)(n - 1)}{n - p - 1} \right]$$

Ekhane n = number of samples, p = number of features.

Intuition: R² feature barale automatically barte thake, even jodi feature useless hoy. Adjusted R² eita penalize kore — useless feature add korle Adjusted R² kome jete pare. **When to use:** Multiple features thakle, feature selection compare korte.

5.6 Comparison Table

Metric	Outlier Sensitivity	Interpretability	Unit
MAE	Kom	High	Original unit
MSE	High	Medium	Squared unit
RMSE	High	High	Original unit
R ²	Medium	High (0-1 scale)	Unitless
Adjusted R ²	Medium	High (feature-penalized)	Unitless

```
from sklearn.metrics import mean_absolute_error, mean_squared_error, r2_score
import numpy as np

mae = mean_absolute_error(y_test, y_pred)
mse = mean_squared_error(y_test, y_pred)
rmse = np.sqrt(mse)
r2 = r2_score(y_test, y_pred)

n, p = X_test.shape[0], X_test.shape[1]
adj_r2 = 1 - ((1 - r2) * (n - 1)) / (n - p - 1)
```

5.7 Summary

- MAE = average absolute error (outlier-friendly)
- MSE/RMSE = boro error-ke beshi penalize kore
- R² = variance explain kora percent
- Adjusted R² = feature count-er jonno penalized R²

5.8 Common Mistakes

- Sudhu R² dekhe model “good” bole declare kora — RMSE/MAE o dekha uchit.
- Feature add korar por Adjusted R² na check kore R² e overconfident hoye jawa.
- Different scale-er dataset e RMSE compare kora (normalize na kore).

5.9 Interview Questions

1. MAE ar RMSE-r moddhe difference ki?
2. R² negative hote pare kokhon?
3. Adjusted R², R² theke keno better multiple feature thakle?
4. Outlier thakle konta metric use korba?

5.10 Quick Revision

MAE (simple avg) < RMSE (penalizes big error) | R^2 (variance explained) | Adjusted R^2 (feature-penalized R^2)

6 Chapter 6: Classification

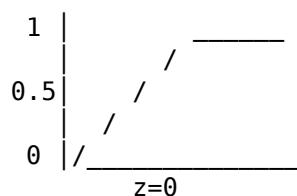
6.1 6.1 Logistic Regression

Intuition: Naam-e “Regression” thakleo eta Classification algorithm — probability predict kore (0 theke 1), tarpor threshold diye class decide kore.

Real-life example: Email Spam(1)/Not Spam(0) predict kora.

Sigmoid Function: Linear output-ke 0-1 range e convert kore:

$$\sigma(z) = \frac{1}{1 + e^{-z}}, \quad z = \beta_0 + \beta_1 x_1 + \dots + \beta_n x_n$$



Odds: Event ghotar probability / na ghotar probability = $\frac{p}{1-p}$

Log Odds (Logit): $\ln\left(\frac{p}{1-p}\right) = z$ — eta-i actually linear equation-er sathe equal.

Binary Cross Entropy (Loss Function):

$$J(\beta) = -\frac{1}{n} \sum_{i=1}^n [y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i)]$$

Decision Boundary: Default threshold 0.5 — probability > 0.5 hole Class 1, na hole Class 0.

Advantages: Fast, interpretable, probability output dey. **Disadvantages:** Non-linear relationship bujhte pare na (feature engineering lage).

Hyperparameters: C (inverse regularization strength), penalty (l1/l2)

```
from sklearn.linear_model import LogisticRegression
model = LogisticRegression(C=1.0)
model.fit(X_train, y_train)
y_pred = model.predict(X_test)
y_prob = model.predict_proba(X_test)
```

Interview Questions: Sigmoid keno use kora hoy Linear Regression-er upor? Log Odds ki? Cross Entropy keno MSE-r bodole use kora hoy classification e? **Common Mistakes:** Threshold shobshomoy 0.5 fix mone kora — imbalanced data e threshold tune korte hoy.

6.2 6.2 Decision Tree (Classification)

Intuition: Feature diye bar bar question kore data ke pure (single-class) group e bhag kora.

Splitting Criteria:

Entropy (impurity measure):

$$Entropy = - \sum_{i=1}^c p_i \log_2(p_i)$$

Entropy 0 mane pure node (shob same class), 1 mane maximum impurity (50-50 split, 2 class case-e).

Information Gain:

$$IG = Entropy(parent) - \sum \frac{n_{child}}{n_{parent}} \times Entropy(child)$$

Jei feature diye split korle Information Gain sobcheye beshi, shei feature-e split kora hoy.

Gini Index (alternative to entropy, computationally faster):

$$Gini = 1 - \sum_{i=1}^c p_i^2$$

Complete Splitting Process Example:

Dataset: 10 samples (6 Yes, 4 No)

Entropy(parent) = $-(0.6 \log_2 0.6 + 0.4 \log_2 0.4) = 0.971$

Split by "Age > 30?":

Left (Age>30): 5 samples (5 Yes, 0 No) → Entropy = 0

Right (Age≤30): 5 samples (1 Yes, 4 No) → Entropy = 0.722

Weighted Entropy = $(5/10)*0 + (5/10)*0.722 = 0.361$

Information Gain = $0.971 - 0.361 = 0.61$ ← highest gain feature bacha hoy

Advantages: Interpretable, non-linear relationship shikhte pare, scaling lage na. **Disadvantages:** Overfit hoya jay shohoje.

Hyperparameters: max_depth, criterion (gini/entropy), min_samples_split

```
from sklearn.tree import DecisionTreeClassifier
model = DecisionTreeClassifier(criterion='gini', max_depth=5)
model.fit(X_train, y_train)
```

Interview Questions: Entropy vs Gini — kokhon konta use korba? Information Gain kivabe calculate hoy? **Common Mistakes:** Pruning na kora — tree onek deep hoye giye overfit kore.

6.3 Random Forest (Classification)

Intuition: Onek Decision Tree banano hoy (bagging + random feature), protita tree ekta “vote” dey, majority vote-e final class decide hoy.

Advantages: High accuracy, overfitting kom, robust to noise. **Disadvantages:** Interpretability kome, computationally heavy.

```
from sklearn.ensemble import RandomForestClassifier
model = RandomForestClassifier(n_estimators=200, max_depth=10)
model.fit(X_train, y_train)
```

Interview Questions: Random Forest e “Out of Bag” (OOB) score ki?

6.4 6.4 K-Nearest Neighbors (KNN)

Intuition: “Tumi tomar cheye-o judge hobe tomar neighbor-der diye” — new point-er k-ta nearest neighbor dekhe, majority class-e assign kora hoy. **Lazy learner** — training e kichu shikhe na, shudhu data store kore rakhe.

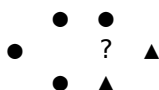
Real-life example: Notun customer ke segment korte, tar kache-kachi thaka existing customer-der pattern dekhe.

Distance Formula (Euclidean):

$$d(p, q) = \sqrt{\sum_{i=1}^n (p_i - q_i)^2}$$

Class A: ●

Class B: ▲



k=3 nearest → 2 ● + 1 ▲ → predict Class A

Advantages: Simple, no training time, non-linear boundary shikhte pare. **Disadvantages:** Prediction time slow (boro dataset e), feature scaling must, curse of dimensionality.

Hyperparameters: n_neighbors (k), metric (distance type)

```
from sklearn.neighbors import KNeighborsClassifier
from sklearn.preprocessing import StandardScaler
```

```
scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)
```

```
model = KNeighborsClassifier(n_neighbors=5)
model.fit(X_train_scaled, y_train)
```

Interview Questions: K choto hole ki hoy? K boro hole ki hoy? (Choto k = overfit/noisy, Boro k = underfit/smooth). Feature scaling KNN e keno mandatory? **Common Mistakes:** Scaling na kora, even K value use kora (tie problem hote pare 2-class e).

6.5 6.5 Naive Bayes

Intuition: Bayes’ Theorem use kore probability calculate kora, “naive” assumption niye j shob feature nijeder moddhe independent (real-e shobshomoy true na hoyeo valo kaj kore).

Formula (Bayes’ Theorem):

$$P(y|X) = \frac{P(X|y) \cdot P(y)}{P(X)}$$

Real-life example: Email spam detection — word frequency diye “spam” probability calculate kora (classic use-case).

Advantages: Khub fast, kom data-teo valo kaj kore, text classification e strong. **Disadvantages:** “Independence assumption” real world e shomoy shomoy fail kore.

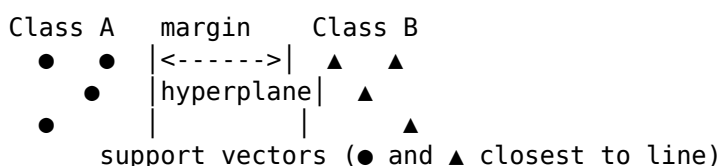
```
from sklearn.naive_bayes import GaussianNB
model = GaussianNB()
model.fit(X_train, y_train)
```

Interview Questions: “Naive” keno bola hoy? Kokhon Naive Bayes valo perform kore?

6.6 Support Vector Machine (SVM)

Intuition: Data ke duita class e alada korar jonno emon ekta line/plane khoja hoy jeta duita class-er moddhe **maximum margin** rakhe.

Hyperplane: Jei line/plane data ke classes e bhag kore. **Margin:** Hyperplane theke closest point porjonto-r distance — SVM eita **maximize** korte chay. **Support Vector:** Hyperplane-er sobcheye kache thaka point gula, jara margin decide kore.



Kernels: Non-linear data ke higher dimension e map kore, jate linearly separable hoy.

Kernel	Use case
Linear	Data already linearly separable
RBF (Radial Basis)	Non-linear, complex boundary
Polynomial	Curved boundary

Advantages: High dimensional data e valo kaj kore, outlier-e robust (margin-based). **Disadvantages:** Boro dataset e slow, kernel/hyperparameter tuning complex.

Hyperparameters: C (regularization), kernel, gamma

```
from sklearn.svm import SVC
model = SVC(kernel='rbf', C=1.0, gamma='scale')
model.fit(X_train_scaled, y_train)
```

Interview Questions: Support Vector ki? C parameter barale ki hoy? Kernel trick ki? **Common Mistakes:** Scaling na kora (SVM distance-based, must scale), boro dataset e linear kernel na try kore direct RBF use kora (slow hoye jay).

6.7 Summary Table

Algorithm	Type	Speed	Interpretability
Logistic Regression	Linear	Fast	High
Decision Tree	Rule-based	Fast	High
Random Forest	Ensemble	Medium	Medium
KNN	Instance-based	Slow(predict)	Medium
Naive Bayes	Probabilistic	Very Fast	Medium
SVM	Margin-based	Slow(train)	Low

6.8 Common Mistakes (Overall)

- Feature scaling na kora KNN/SVM/Logistic Regression-e.
- Imbalanced dataset e sudhu Accuracy dekhe model valo bole dhore neya.
- Decision Tree-e pruning na kora.

6.9 Interview Questions (Overall)

1. Entropy ar Gini Index-er moddhe difference ki?
2. KNN-ke “lazy learner” kano bola hoy?
3. SVM-e kernel trick ki, keno lage?
4. Naive Bayes-er “naive” assumption ki?

6.10 Quick Revision

Logistic → sigmoid+probability | Decision Tree → entropy/gini split | Random Forest → many trees vote | KNN → nearest neighbor vote | Naive Bayes → Bayes theorem | SVM → max margin hyperplane

7 Chapter 7: Classification Metrics

7.1 7.1 Confusion Matrix

Confusion Matrix hoche ekta table jeta actual vs predicted class-er comparison dekhay.

	Predicted Positive	Predicted Negative
Actual Positive	TP (True Positive)	FN (False Negative)
Actual Negative	FP (False Positive)	TN (True Negative)

Real example: Disease detection — TP mane roga-ke roga bola hoise (thik), FN mane roga-ke healthy bola hoise (khub dangerous mistake!), FP mane healthy manush ke roga bola hoise (false alarm), TN mane healthy-ke healthy bola hoise (thik).

7.2 7.2 Accuracy

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

Total koto percent prediction thik hoise. **Problem:** Imbalanced dataset e misleading — 95% “No Disease” data hole shudhu “No” bolleo 95% accuracy pawa jabe!

7.3 7.3 Precision

$$Precision = \frac{TP}{TP + FP}$$

Jokhon model “Positive” bole, koto percent shomoy shotti positive. **Use when:** False Positive costly (jemon spam detection — normal email spam bolle problem).

7.4 7.4 Recall (Sensitivity)

$$Recall = \frac{TP}{TP + FN}$$

Actual positive-der moddhe koto percent model dhorte parlo. **Use when:** False Negative costly (jemon disease detection — roga miss kora dangerous).

7.5 7.5 Specificity

$$Specificity = \frac{TN}{TN + FP}$$

Actual negative-der moddhe koto percent thik-thak negative bola hoise.

7.6 7.6 F1 Score

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

Precision ar Recall-er **harmonic mean** — jorkhon duitar balance dorkar, ba dataset imbalanced.

Important Formula Box Accuracy = overall correctness | Precision = positive prediction-er quality | Recall = actual positive koto dhora gelo | F1 = Precision-Recall balance

7.7 7.7 Confusion Matrix Calculation Example

Actual: [1,1,0,1,0,0,1,0]
Predicted: [1,0,0,1,0,1,1,0]

TP=3 (actual1,pred1: idx0,3,6)

FN=1 (actual1,pred0: idx1)

FP=1 (actual0,pred1: idx5)

TN=3 (actual0,pred0: idx2,4,7)

Accuracy = (3+3)/8 = 0.75

Precision = 3/(3+1) = 0.75

Recall = 3/(3+1) = 0.75

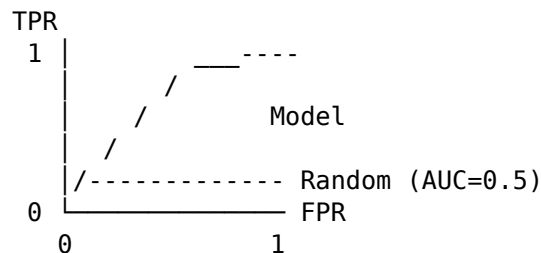
F1 = 2*(0.75*0.75)/(0.75+0.75) = 0.75

7.8 7.8 ROC Curve & AUC

ROC Curve: Different threshold e True Positive Rate (Recall) vs False Positive Rate plot kora hoy.

$$FPR = \frac{FP}{FP + TN}$$

AUC (Area Under Curve): ROC curve-er niche-r area — model koto valo positive/negative alada korte pare, 1.0 = perfect, 0.5 = random guessing.



7.9 7.9 Precision-Recall Curve

Different threshold e Precision vs Recall plot kora hoy — highly imbalanced dataset e ROC-AUC theke eta beshi meaningful hote pare.

7.10 Comparison Table

Metric	Formula focus	Best jikhon
Accuracy	Overall correctness	Balanced dataset
Precision	FP minimize	False Positive costly (spam)
Recall	FN minimize	False Negative costly (disease)
F1 Score	Precision+Recall balance	Imbalanced dataset
AUC-ROC	Overall separability	Model comparison

```
from sklearn.metrics import (confusion_matrix, accuracy_score, precision_score,
                             recall_score, f1_score, roc_auc_score, roc_curve)

cm = confusion_matrix(y_test, y_pred)
acc = accuracy_score(y_test, y_pred)
prec = precision_score(y_test, y_pred)
rec = recall_score(y_test, y_pred)
f1 = f1_score(y_test, y_pred)
auc = roc_auc_score(y_test, y_prob[:,1])
```

7.11 Summary

- Confusion Matrix holo shob metric-er base.
- Accuracy imbalanced data-e misleading, tokhon F1/Precision/Recall use koro.
- AUC-ROC model comparison-er jonno strong tool.

7.12 Common Mistakes

- Imbalanced dataset e sudhu Accuracy report kora.
- Precision-Recall tradeoff na bujhe, threshold change korle duitar upor effect na bujha.
- Multi-class problem e binary metric formula directly apply korar chesta kora (averaging method lage: macro/micro/weighted).

7.13 Interview Questions

1. Precision ar Recall-er moddhe tradeoff ki?
2. Imbalanced dataset e konta metric use korba ar keno?
3. AUC = 0.5 mane ki?
4. Confusion Matrix theke F1 score kivabe calculate korba?
5. Type I ar Type II error ki? (FP = Type I, FN = Type II)

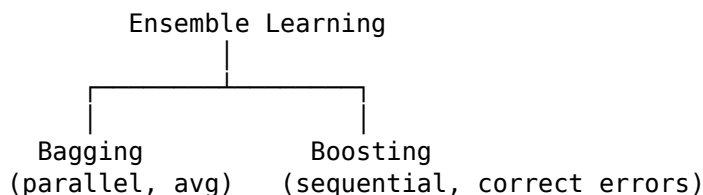
7.14 Quick Revision

Accuracy = overall | Precision = FP kome | Recall = FN kome | F1 = balance | AUC = separability

8 Chapter 8: Ensemble Learning

8.1 8.1 Ensemble Learning Ki?

Ensemble Learning mane multiple model-ke combine kore, single model theke better prediction pawa. **Analogy:** Ekjon doctor-er opinion neya theke, 10 jon doctor-er opinion niye majority-decision neya beshi reliable.



8.2 8.2 Bagging (Bootstrap Aggregating)

Intuition: Original data theke random sample (with replacement) niye multiple model **parallel** e train kora hoy, tarpor shob-er prediction average/vote kora hoy. Eta **variance** kome.

Example: Random Forest — Bagging-er sobcheye famous example.

8.3 8.3 Boosting

Intuition: Model gula **sequentially** train hoy — proti model ager model-er “vul” (error) theke shikhe, correct korar chesta kore. Eta **bias** kome.

```
Model1 → Error1
      ↓ (focus on wrong predictions)
Model2 → Error2 (kom)
      ↓
Model3 → Error3 (aro kom)
      ↓
Final = Weighted combination
```

8.4 8.4 Random Forest (Recap)

Bagging + Random Feature Selection — multiple Decision Tree, prediction average/vote.

8.5 8.5 AdaBoost (Adaptive Boosting)

Intuition: Proti round e jei sample gula “misclassified” hoy, tader **weight barano** hoy, jate next model shei hard sample-gulor upor beshi focus kore.

Advantages: Simple, interpretable weak-learner combination. **Disadvantages:** Outlier/noise-e sensitive (beshi weight paye jay).

```
from sklearn.ensemble import AdaBoostClassifier
model = AdaBoostClassifier(n_estimators=100, learning_rate=1.0)
model.fit(X_train, y_train)
```

8.6 8.6 Gradient Boosting

Intuition: Proti notun model, ager model-er **residual error** predict korar chesta kore (gradient descent-er moto approach diye loss minimize kora).

Advantages: Very high accuracy. **Disadvantages:** Slow training, overfitting-prone jodi tune na kora hoy.

```
from sklearn.ensemble import GradientBoostingClassifier
model = GradientBoostingClassifier(n_estimators=100, learning_rate=0.1, max_depth=3)
model.fit(X_train, y_train)
```

8.7 8.7 XGBoost (Extreme Gradient Boosting)

Intuition: Gradient Boosting-er optimized, regularized, parallel version — Kaggle competition-e sobcheye popular algorithm.

Advantages: Fast, regularization built-in, missing value handle kore, high performance.

```
import xgboost as xgb
model = xgb.XGBClassifier(n_estimators=200, learning_rate=0.1, max_depth=6)
model.fit(X_train, y_train)
```

8.8 8.8 LightGBM

Intuition: Microsoft-er banana, “leaf-wise” tree growth use kore (level-wise na), tai onek fast, boro dataset-e efficient.

```
import lightgbm as lgb
model = lgb.LGBMClassifier(n_estimators=200, learning_rate=0.1)
model.fit(X_train, y_train)
```

8.9 8.9 CatBoost

Intuition: Categorical feature ke automatically handle korte pare (one-hot encoding lage na), Yandex-er banana.

```
from catboost import CatBoostClassifier
model = CatBoostClassifier(iterations=200, learning_rate=0.1, verbose=0)
model.fit(X_train, y_train)
```

8.10 8.10 Comparison Table

Algorithm	Type	Speed	Categorical Handling	Best For
Random Forest	Bagging	Medium	No (encode lage)	General purpose
AdaBoost	Boosting	Medium	No	Simple boosting
Gradient Boosting	Boosting	Slow	No	High accuracy
XGBoost	Boosting	Fast	No (encode lage)	Kaggle, tabular data
LightGBM	Boosting	Very Fast	Partial	Boro dataset
CatBoost	Boosting	Fast	Yes (built-in)	Categorical-heavy data

8.11 Summary

- Bagging = parallel models, variance kome (Random Forest)
- Boosting = sequential models, bias kome (AdaBoost, GBM, XGBoost, LightGBM, CatBoost)
- Boosting algorithms shadaronoto Bagging theke beshi accurate, kintu overfitting-prone hole tune korte hoy.

8.12 Common Mistakes

- Boosting model-e learning_rate khub boro rakha (overfitting/instability).
- Bagging ar Boosting-ke same jinis mone kora.
- Categorical variable XGBoost e directly na encode kore feed kora (error dey).

8.13 Interview Questions

1. Bagging ar Boosting-er moddhe difference ki?
2. XGBoost ke keno “extreme” bola hoy?
3. Random Forest ar Gradient Boosting-er moddhe main difference ki?
4. LightGBM keno fast?
5. Overfitting rodh korte Boosting-e ki ki hyperparameter tune korba?

8.14 Quick Revision

Bagging(parallel,variance↓): Random Forest | Boosting(sequential,bias↓): AdaBoost, GBM, XGBoost, LightGBM, CatBoost

9 Chapter 9: Hyperparameter Tuning

Hyperparameter hoche shei setting ja model **train howar age** manually set kora hoy (jemon Random Forest-e `n_estimators`), model nijei eta data theke shikhe na.

9.1 9.1 GridSearchCV

Intuition: Shob possible hyperparameter combination systematically try kora hoy (brute force), cross-validation diye best combination bacha hoy.

```
from sklearn.model_selection import GridSearchCV
from sklearn.ensemble import RandomForestClassifier

params = {'n_estimators': [100, 200], 'max_depth': [5, 10, None]}
grid = GridSearchCV(RandomForestClassifier(), params, cv=5, scoring='accuracy')
grid.fit(X_train, y_train)
print(grid.best_params_)
```

Advantages: Guaranteed best combination (given grid). **Disadvantages:** Slow — combination barle exponentially time barbe.

9.2 9.2 RandomizedSearchCV

Intuition: Shob combination na try kore, random-e kichu combination try kora hoy — fast, kintu best na-o pete pare, “good enough” pay shohoje.

```
from sklearn.model_selection import RandomizedSearchCV
random_search = RandomizedSearchCV(RandomForestClassifier(), params, n_iter=10, cv=5)
random_search.fit(X_train, y_train)
```

9.3 9.3 Bayesian Optimization

Intuition: Age-r try-gulor result dekhe, “intelligent guess” diye porer combination choose kora hoy (probabilistic model use kore) — GridSearch/RandomSearch theke onek smart ar fast.

```
# using scikit-optimize
from skopt import BayesSearchCV
opt = BayesSearchCV(RandomForestClassifier(), {'n_estimators': (50, 300), 'max_depth': (3, 20)}, n_i
opt.fit(X_train, y_train)
```

9.4 9.4 Early Stopping

Intuition: Boosting/Neural Network train korar shomoy, validation error kome kome ek shomoy barte shuru korle (overfitting shuru), training oikhanei theme jay — extra unnecessary training bondho hoy.

```
model = xgb.XGBClassifier(n_estimators=1000)
model.fit(X_train, y_train, eval_set=[(X_val, y_val)],
        early_stopping_rounds=20, verbose=False)
```

9.5 Comparison Table

Method	Speed	Guarantee	Best For
GridSearchCV	Slow	Best in grid	Choto search space
RandomizedSearchCV	Fast	No guarantee	Boro search space
Bayesian Optimization	Medium	Smart search	Complex, expensive model
Early Stopping	N/A	Overfitting rodh	Boosting/Neural Net

9.6 Summary

Hyperparameter tuning model-er optimal configuration khoja — Grid (exhaustive), Random (fast sampling), Bayesian (smart), Early Stopping (training-time control).

9.7 Common Mistakes

- Tuning-er shomoy Test set use kora (Validation set/CV use kora uchit).
- Search space onek boro rekhe GridSearch use kora — onek shomoy lagbe.

9.8 Interview Questions

1. GridSearch ar RandomSearch-er moddhe difference ki?
2. Bayesian Optimization keno beshi efficient?
3. Early Stopping kivabe overfitting rodh kore?

9.9 Quick Revision

GridSearch(exhaustive,slow) | RandomSearch(sample,fast) | Bayesian(smart) | Early Stopping(training control)

10 Chapter 10: Cross Validation

Cross Validation hoche model-er performance ke reliably estimate korar technique — sudhu ekbar train/test split korle result “lucky” hote pare, tai bar bar different split e test kora hoy.

10.1 10.1 Hold-Out Validation

Intuition: Simple train/test split (e.g. 80/20) — ekbar-i split kora hoy.

Disadvantages: Result split-er upor depend kore, “luck” factor thake.

10.2 10.2 K-Fold Cross Validation

Intuition: Data ke K-ta equal part (fold)-e bhag kora hoy, protibar 1 fold test set hishebe rakha hoy, baki K-1 fold train — eta K bar repeat kora hoy, tarpor average result neya hoy.

```
Fold1 Fold2 Fold3 Fold4 Fold5
[Test][Train][Train][Train][Train] → Run 1
[Train][Test][Train][Train][Train] → Run 2
[Train][Train][Test][Train][Train] → Run 3
...
Final Score = Average of all runs

from sklearn.model_selection import cross_val_score
scores = cross_val_score(model, X, y, cv=5)
print(scores.mean())
```

10.3 10.3 Stratified K-Fold

Intuition: Normal K-Fold e class distribution unequal hote pare protita fold-e. Stratified K-Fold protita fold-e original class-ratio maintain kore — **imbalanced classification e mandatory**.

```
from sklearn.model_selection import StratifiedKFold
skf = StratifiedKFold(n_splits=5, shuffle=True, random_state=42)
```

10.4 10.4 Leave-One-Out (LOOCV)

Intuition: K = n (total sample count) — protibar 1-ta sample test, baki shob train. Extremely thorough kintu extremely slow.

10.5 Comparison Table

Method	Kokhon Use Korba
Hold-Out	Boro dataset, fast prototype
K-Fold	General purpose, medium dataset
Stratified K-Fold	Classification, imbalanced data
LOOCV	Khub choto dataset

10.6 Summary

Cross Validation model-er reliable evaluation dey — K-Fold sobcheye commonly use hoy, Classification e Stratified K-Fold better.

10.7 Common Mistakes

- Imbalanced classification e normal K-Fold use kora (Stratified use kora uchit).
- Cross Validation-er por o test set-e final evaluation na kora.
- Data leakage — feature scaling/engineering CV fold split-er age kore fela.

10.8 Interview Questions

1. K-Fold ar Stratified K-Fold-er difference ki?
2. K barale ki hoy (bias-variance context e)?
3. LOOCV keno slow, kokhon use korba?

10.9 Quick Revision

Hold-Out(simple) | K-Fold(average of K) | Stratified(class-ratio maintain) | LOOCV($K=n$, thorough but slow)

11 Chapter 11: Feature Engineering

11.1 11.1 Missing Values

Techniques: - **Drop:** Row/column drop kora (jodi missing kom hoy) - **Mean/Median Imputation:** Numerical column-e - **Mode Imputation:** Categorical column-e - **KNN Imputer:** Nearby sample-er value diye fill kora

```
from sklearn.impute import SimpleImputer
imputer = SimpleImputer(strategy='median')
X_filled = imputer.fit_transform(X)
```

11.2 11.2 Outlier Handling

Detection: Boxplot, Z-score, IQR method.

$$IQR = Q3 - Q1, \quad \text{Outlier} : x < Q1 - 1.5 \times IQR \text{ or } x > Q3 + 1.5 \times IQR$$

Handling: Remove, Cap (clip to boundary), ba Transform (log transform).

11.3 11.3 Feature Scaling

Standardization (Z-score):

$$x' = \frac{x - \mu}{\sigma}$$

Mean=0, Std=1 kore dey. Distance-based algorithm (KNN, SVM) ar gradient-based (Linear/Logistic Regression, Neural Net) e must.

Normalization (Min-Max Scaling):

$$x' = \frac{x - x_{min}}{x_{max} - x_{min}}$$

Value ke 0-1 range e niye ashe.

```
from sklearn.preprocessing import StandardScaler, MinMaxScaler
scaler = StandardScaler() # or MinMaxScaler()
X_scaled = scaler.fit_transform(X)
```

11.4 11.4 Label Encoding

Intuition: Categorical value-ke number-e convert kora (Red=0, Green=1, Blue=2). **Careful:** Ordinal relationship na thakle model bhul order assume korte pare.

```
from sklearn.preprocessing import LabelEncoder
le = LabelEncoder()
y_encoded = le.fit_transform(y)
```

11.5 11.5 One-Hot Encoding

Intuition: Protita category-r jonno alada binary column banano — no false ordinal relationship.

```
import pandas as pd
df_encoded = pd.get_dummies(df, columns=['color'])
```

11.6 Target Encoding

Intuition: Category ke target variable-er mean value diye replace kora — high-cardinality categorical feature-e useful, kintu **data leakage** hote pare (careful cross-validation lage).

11.7 Feature Selection

Techniques: - **Filter method:** Correlation, Chi-square test - **Wrapper method:** Recursive Feature Elimination (RFE) - **Embedded method:** Lasso, Tree-based feature importance

```
from sklearn.feature_selection import RFE
rfe = RFE(estimator=RandomForestClassifier(), n_features_to_select=5)
rfe.fit(X_train, y_train)
```

11.8 Comparison Table

Technique	Type	Kokhon Use
Label Encoding	Ordinal category	Order matter kore (Low/Medium/High)
One-Hot Encoding	Nominal category	Order matter kore na (Color)
Target Encoding	High-cardinality	On ek unique category
Standardization	Scaling	Gradient/Distance-based model
Normalization	Scaling	Neural Network, bounded range lage

11.9 Summary

Feature Engineering-i onektai model-er accuracy determine kore — missing value, outlier, scaling, encoding shob thik-thak korle simple algorithm-o valo result dey.

11.10 Common Mistakes

- One-Hot Encoding-er bodole Label Encoding use kora nominal category-r jonno (fake order create hoy).
- Train/Test split-er age scaling fit kora (data leakage).
- High-cardinality categorical feature One-Hot Encode kore column explosion kora.

11.11 Interview Questions

1. Standardization ar Normalization-er difference ki?
2. Label Encoding kokhon problematic hote pare?
3. Data leakage feature engineering-e kivabe hoy?
4. One-Hot Encoding-er “dummy variable trap” ki?

11.12 Quick Revision

Missing→impute | Outlier→IQR/cap | Scaling→Standard(mean0,std1)/MinMax(0-1) |
Encoding→Label(ordinal)/OneHot(nominal)/Target(high-cardinality)

12 Chapter 12: Complete Machine Learning Pipeline

Ei chapter e amra ekta complete project banabo — start theke end porjonto, ager shob chapter-er concept ekshathe apply kore.

Problem: House Price Prediction (Regression)

12.1 12.1 Data Loading

```
import pandas as pd
df = pd.read_csv("house_data.csv")
print(df.head())
print(df.shape)
```

12.2 12.2 EDA

```
df.info()
df.describe()

import matplotlib.pyplot as plt
import seaborn as sns

sns.histplot(df['price'])
plt.show()

sns.heatmap(df.corr(numeric_only=True), annot=True)
plt.show()
```

12.3 12.3 Data Cleaning

```
df = df.drop_duplicates()
df['bedrooms'] = df['bedrooms'].fillna(df['bedrooms'].median())

Q1 = df['price'].quantile(0.25)
Q3 = df['price'].quantile(0.75)
IQR = Q3 - Q1
df = df[(df['price'] >= Q1 - 1.5*IQR) & (df['price'] <= Q3 + 1.5*IQR)]
```

12.4 12.4 Feature Engineering

```
df['house_age'] = 2026 - df['year_built']
df['price_per_sqft'] = df['price'] / df['area_sqft']

df = pd.get_dummies(df, columns=['location'], drop_first=True)
```

12.5 12.5 Feature Selection & Split

```
X = df.drop(columns=['price'])
y = df['price']
```

```
from sklearn.model_selection import train_test_split
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)
```

12.6 12.6 Scaling

```
from sklearn.preprocessing import StandardScaler
scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)
```

12.7 12.7 Model Building (Multiple Model Comparison)

```
from sklearn.linear_model import LinearRegression, Ridge
from sklearn.ensemble import RandomForestRegressor
from xgboost import XGBRegressor
from sklearn.metrics import r2_score, mean_squared_error
import numpy as np

models = {
    "Linear Regression": LinearRegression(),
    "Ridge": Ridge(alpha=1.0),
    "Random Forest": RandomForestRegressor(n_estimators=200, random_state=42),
    "XGBoost": XGBRegressor(n_estimators=200, learning_rate=0.1)
}

results = {}
for name, model in models.items():
    model.fit(X_train_scaled, y_train)
    pred = model.predict(X_test_scaled)
    results[name] = {
        "R2": r2_score(y_test, pred),
        "RMSE": np.sqrt(mean_squared_error(y_test, pred))
    }

print(pd.DataFrame(results).T)
```

12.8 12.8 Hyperparameter Tuning (Best Model)

```
from sklearn.model_selection import GridSearchCV

params = {'n_estimators': [100, 200, 300], 'max_depth': [5, 10, None]}
grid = GridSearchCV(RandomForestRegressor(random_state=42), params, cv=5, scoring='r2')
grid.fit(X_train_scaled, y_train)

print(grid.best_params_)
best_model = grid.best_estimator_
```


12.9 12.9 Final Evaluation

```
final_pred = best_model.predict(X_test_scaled)
print("Final R2:", r2_score(y_test, final_pred))
print("Final RMSE:", np.sqrt(mean_squared_error(y_test, final_pred)))
```

12.10 12.10 Pipeline Flowchart (Full Recap)

Load Data → EDA → Clean → Feature Engineer → Encode → Split
→ Scale → Train Multiple Models → Compare → Tune Best Model
→ Final Evaluate → Save/Deploy

12.11 Summary

Ekta real ML project always structured pipeline follow kore — kono step skip korle result unreliable hobe.

12.12 Common Mistakes

- Scaler ke fit_transform test set-e o apply kora (test-e sudhu transform use korte hobe).
- Model comparison na kore first algorithm-i final dhore neya.
- Hyperparameter tuning-er por abar notun test set-e evaluate na kora.

12.13 Interview Questions

1. Kivabe decide korba kon model final use korba multiple model try korar por?
2. Scaler-e fit_transform ar transform-er difference ki, ar keno test set e sudhu transform use kora hoy?
3. Pipeline object (sklearn.pipeline.Pipeline) use korar advantage ki?

12.14 Quick Revision

Load → EDA → Clean → Feature Engineer → Split → Scale → Train(multiple) → Compare → Tune → Evaluate → Deploy

13 Chapter 13: Interview Questions (100+ with Answers)

13.1 A. AI/ML Basics (Q1-10)

Q1. AI, ML, DL-er difference ki? AI = broad goal (intelligent machine), ML = data theke shikhe prediction dewa technique, DL = neural network based ML subset.

Q2. Supervised ar Unsupervised Learning-er difference ki? Supervised e labeled data thake (input+output), Unsupervised e sudhu input thake, model nijei pattern/cluster khoje.

Q3. Reinforcement Learning ki? Agent environment-e action nay, reward/punishment pay, ei feedback diye policy shikhe — jemon game-playing AI.

Q4. Overfitting ki? Model training data khub bhalo shikhe felay, kintu notun/unseen data e kharap perform kore — model “memorize” kore felse, “generalize” kore nai.

Q5. Underfitting ki? Model shothik pattern-i shikhte parche na — training ar test dutoteo kharap perform kore, model onek simple.

Q6. Bias-Variance Tradeoff ki? Bias = model-er simplifying assumption-er error (underfitting related), Variance = data change-e model koto sensitive (overfitting related). Duita-r balance-e best model pawa jay.

Q7. Curse of Dimensionality ki? Feature (dimension) barle data sparse hoye jay, distance-based algorithm-er performance kharap hoye jay.

Q8. Parametric vs Non-Parametric model? Parametric = fixed number of parameters (Linear Regression), Non-Parametric = data-er shathe complexity barte pare (KNN, Decision Tree).

Q9. Generative vs Discriminative model? Generative = $P(X,y)$ shikhe, notun data generate korte pare (Naive Bayes), Discriminative = $P(y|X)$ shikhe, direct decision boundary shikhe (Logistic Regression, SVM).

Q10. Machine Learning-er 3 main type ki ki? Supervised, Unsupervised, Reinforcement Learning.

13.2 B. Data Preprocessing (Q11-25)

Q11. Missing value handle korar 3-ta way bolo. Drop, Mean/Median/Mode imputation, KNN Imputer.

Q12. Outlier detect korar way ki? Boxplot, Z-score (>3 std), IQR method.

Q13. Standardization ar Normalization-er difference? Standardization = mean 0, std 1; Normalization = 0-1 range e scale kora.

Q14. One-Hot Encoding kokhon use korba? Nominal categorical data (order nai), jemon Color, City.

Q15. Label Encoding kokhon risky? Nominal data-e use korle model fake order assume kore felte pare.

Q16. Data Leakage ki? Test/future information accidentally training process-e chole ashle — model unrealistically valo perform kore mone hoy but production e fail kore.

Q17. Class Imbalance handle korar way ki? Oversampling (SMOTE), Undersampling, Class weight adjustment.

Q18. SMOTE ki? Synthetic Minority Oversampling Technique — minority class-er notun synthetic sample generate kore balance kora.

Q19. Feature Scaling kon algorithm-e mandatory? KNN, SVM, Logistic/Linear Regression (gradient descent), Neural Network, PCA.

Q20. Feature Scaling kon algorithm-e lage na? Decision Tree, Random Forest, XGBoost (tree-based, split threshold-based).

Q21. Multicollinearity ki? Duita ba beshi feature nijeder moddhe highly correlated hole coefficient estimation unstable hoye jay.

Q22. Multicollinearity detect korar way? VIF (Variance Inflation Factor), Correlation matrix.

Q23. Target Encoding-er risk ki? Data leakage — proper cross-validation na korle overfit hoye jay.

Q24. Dimensionality Reduction ki, keno lage? Feature-er sonkhya kombano, jemon PCA — visualization, speed, curse of dimensionality solve korar jonno.

Q25. PCA ki? Principal Component Analysis — data-r variance sobcheye beshi capture kore emon notun (uncorrelated) axis-e project kora.

13.3 C. Regression (Q26-40)

Q26. Linear Regression-er assumptions ki ki? Linearity, No multicollinearity, Homoscedasticity, Normal residuals.

Q27. Gradient Descent ki? Cost function minimize korar iterative optimization algorithm — proti step e gradient-er opposite direction e move kore.

Q28. Learning Rate barale ki hoy? Boro learning rate e overshoot hote pare (converge na o korte pare), choto hole slow convergence hoy.

Q29. Ridge ar Lasso-r difference? Ridge (L2) coefficient chotto kore kintu 0 kore na; Lasso (L1) coefficient ekdom 0 kore dite pare (feature selection).

Q30. R^2 negative hote pare kokhon? Model jokhon mean-baseline theke o kharap perform kore.

Q31. Heteroscedasticity ki? Residual-er variance constant thake na — Linear Regression-er assumption violate hoy.

Q32. Polynomial Regression e overfitting kivabe rodh korba? Degree kom rakha, Regularization (Ridge/Lasso) apply kora, more data collect kora.

Q33. MAE ar RMSE kokhon different result dey? Outlier thakle — RMSE outlier-ke onek beshi weight dey, MAE dey na.

Q34. Regression e feature importance kivabe ber korba? Coefficient magnitude (scaled data-e), ba tree-based model-e `feature_importances_`.

Q35. Homoscedasticity ki? Error-er variance shob prediction-e roughly constant thaka.

Q36. Decision Tree Regressor-e splitting criteria ki? Variance Reduction / MSE reduction.

Q37. Random Forest Regressor overfitting kore kokhon? `max_depth` unlimited rakhle, `n_estimators` khub kom hole unstable.

Q38. ElasticNet kokhon use korba? Onek correlated feature ache ar feature selection-o lage — Ridge+Lasso duitar benefit lagbe emon case-e.

Q39. Adjusted R^2 kokhon R^2 theke kome jay? Notun feature add korle jodi shei feature model-e meaningfully contribute na kore.

Q40. Log-transformation regression e keno use kora hoy? Skewed target variable-ke normal distribution-er kache anar jonno, heteroscedasticity kombate.

13.4 D. Classification (Q41-60)

Q41. Logistic Regression-e Sigmoid function-er kaj ki? Linear output-ke 0-1 probability range e convert kora.

Q42. Decision Boundary ki? Jei line/surface duita class-ke separate kore.

Q43. Entropy 0 hole ki bujhay? Node completely pure — shob sample same class.

Q44. Gini Index ar Entropy-r moddhe kon-ta fast compute? Gini Index (log calculation lage na).

Q45. KNN-e k=1 rakhle ki hobe? High variance, noise-e very sensitive, overfitting hote pare.

Q46. SVM-e “C” parameter-er kaj ki? Regularization strength — boro C mane kom misclassification tolerate kora (overfit risk), choto C mane beshi tolerate kora (smoother margin).

Q47. Kernel Trick ki? Data-ke higher dimension e explicitly na niye giyeo, higher dimension-er moto separation pawa (mathematical trick, computation save kore).

Q48. Naive Bayes-e “naive” keno bola hoy? Shob feature-ke independent dhore neya hoy, real-e always true hoy na.

Q49. Multi-class Classification-e Logistic Regression kivabe kaj kore? One-vs-Rest (OvR) ba Softmax (Multinomial) approach.

Q50. Decision Tree ar Random Forest-er main difference? Decision Tree = single tree (overfit-prone), Random Forest = many tree-r average/vote (stable).

Q51. Class imbalance-e KNN-er problem ki? Majority class-er dike bias hoya jay (majority neighbor-i beshi thakbe).

Q52. Logistic Regression-e regularization keno lage? Overfitting rodh korte, high-dimensional data-e.

Q53. Binary Cross Entropy-r formula ki? $-\frac{1}{n} \sum [y \log(\hat{y}) + (1 - y) \log(1 - \hat{y})]$

Q54. Hard Margin vs Soft Margin SVM? Hard Margin = kono misclassification allow kore na (noise-e fragile), Soft Margin = kichu misclassification allow kore (C parameter diye control).

Q55. Decision Tree-e “pruning” ki? Overfitting rodh korte tree-r unnecessary branch remove kora (pre-pruning: max_depth set kora, post-pruning: full tree grow kore pore prune kora).

Q56. Probability calibration ki? Model-er predicted probability-ke actual-real-world probability-r sathe align kora (jemon Platt Scaling).

Q57. One-vs-One vs One-vs-Rest? OvR: proti class-er jonno 1 binary classifier (class vs rest); OvO: protijoray class-er jonno 1 classifier.

Q58. KNN-e distance metric hishebe Manhattan kokhon use korba? High dimensional ba grid-like data-e, jekhane diagonal movement meaningful na.

Q59. Naive Bayes-er variant ki ki? GaussianNB (continuous), MultinomialNB (count data/text), BernoulliNB (binary features).

Q60. Decision Tree-e categorical feature-e high cardinality thakle problem ki? Overfitting risk barse, tree biased hote pare beshi-category feature-er dike.

13.5 E. Metrics (Q61-70)

Q61. Confusion Matrix-e TP, FP, TN, FN ki? TP=correct positive, FP=wrong positive predicted, TN=correct negative, FN=wrong negative predicted (missed positive).

Q62. F1 Score keno use kora hoy Accuracy-r bodole? Imbalanced dataset-e Accuracy misleading, F1 Precision-Recall balance kore.

Q63. ROC Curve-e x ar y axis ki? x-axis = False Positive Rate, y-axis = True Positive Rate (Recall).

Q64. AUC = 1.0 mane ki? Model perfect-ly positive ar negative class alada korte parche.

Q65. Type I ar Type II Error ki? Type I = False Positive, Type II = False Negative.

Q66. Precision-Recall Curve kokhon ROC theke better? Highly imbalanced dataset-e (minority class important hole).

Q67. Macro vs Micro F1 score? Macro = protita class-er F1-er simple average, Micro = total TP/FP/FN diye overall calculate kora.

Q68. Log Loss ki? Probability-based classification metric — wrong confident prediction-ke heavily penalize kore.

Q69. Specificity ar Recall-er difference? Recall = actual positive-der koto ta dhorlo, Specificity = actual negative-der koto ta thik dhorlo.

Q70. R² ar Accuracy — same jinis? Na, R² regression metric (variance explained), Accuracy classification metric (correct prediction ratio).

13.6 F. Ensemble & Boosting (Q71-80)

Q71. Bagging ar Boosting-er moddhe main difference? Bagging = parallel, variance kome; Boosting = sequential, bias kome.

Q72. Random Forest-e feature randomness keno lage? Tree-gulor moddhe correlation kombate, jate ensemble beshi diverse hoy.

Q73. XGBoost regularization kivabe kore? L1/L2 regularization built-in loss function-e, tree complexity penalty.

Q74. Gradient Boosting overfitting korle ki korba? learning_rate kombano, max_depth kombano, n_estimators kombano ba early stopping use kora.

Q75. LightGBM-er “leaf-wise” growth ki? Level-wise (shob node ek shathe grow) na kore, sobcheye boro loss-reduction wala leaf-ta age grow kora hoy — fast kintu overfit-prone (max_depth control lage).

Q76. CatBoost-er specialty ki? Categorical feature automatically, efficiently handle kore, one-hot encode korar dorkar nai.

Q77. Stacking ki? Multiple different model-er prediction-ke input hishebe niye, ekta “meta-model” final prediction dey.

Q78. Voting Classifier ki? Multiple model-er prediction-e simple majority vote (hard voting) ba probability average (soft voting).

Q79. AdaBoost-e weak learner kake bola hoy? Shadharon-toh shallow Decision Tree (Decision Stump, depth=1) — random guess theke shamanno valo model.

Q80. Ensemble model interpretability kom keno? Multiple model-er combination hoyar karone individual decision trace kora kothin hoye jay (“black box”).

13.7 G. Model Selection & Tuning (Q81-90)

Q81. Cross Validation keno lage? Model-er performance-ke reliable-vabe estimate korte, single split-er “luck” avoid korte.

Q82. K-Fold-e K=10 rakha standard keno? Bias ar Variance-er moddhe valo balance dey (empirically proven), computationally o feasible.

Q83. GridSearchCV-er downside ki? Search space boro hole exponentially slow hoye jay.

Q84. Early Stopping kivabe kaj kore? Validation loss improve howa bondho hole (patience rounds por), training auto-stop hoye jay.

Q85. Hyperparameter ar Parameter-er difference? Parameter model nijei data theke shikhe (weight, coefficient), Hyperparameter manually set kora hoy training-er age.

Q86. Stratified K-Fold keno classification-e important? Protita fold-e original class ratio maintain rakhe, imbalanced data-e reliable evaluation dey.

Q87. Nested Cross Validation ki? Hyperparameter tuning ar model evaluation-ke completely separate rakhar jonno — outer CV evaluation, inner CV tuning-er jonno.

Q88. Bayesian Optimization GridSearch theke fast keno? Age-r result use kore “smart” next guess dey, shob combination try kore na.

Q89. Model-er “best” configuration select korar shomoy shudhu training score dekha thik na keno? Training score high hole overfitting hote pare — validation/test score-i real generalization bujhay.

Q90. Learning Curve ki? Training size barar shathe train ar validation error-er change plot kora — overfitting/underfitting diagnose korte help kore.

13.8 H. General / Advanced (Q91-100+)

Q91. Deep Learning kokhon classical ML theke better? Onek boro dataset ar unstructured data (image, text, audio) e.

Q92. Batch vs Online Learning? Batch = pura dataset ek shathe diye train, Online = data ashar shathe shathe incrementally update.

Q93. Feature Importance ar Feature Selection same? Na — Feature Importance holo measure (kon feature koto important), Feature Selection holo action (kon feature rakhba/felba).

Q94. Model Drift ki? Shomoyer shathe real-world data distribution change hoye jay, production-e model-er accuracy dheere dheere kome jay.

Q95. A/B Testing ML e kothay use hoy? Notun model deploy korar age, purono ar notun model-er real-world performance compare korte.

Q96. Ensemble-e Diversity keno important? Model-gula jodi shob same vul kore, ensemble-er kono lav nai — diverse model-e combined error kome.

Q97. Imbalanced Dataset-e Random Forest kemon perform kore ar ki korba? Majority class-er dike bias hote pare — `class_weight='balanced'` use kora ba SMOTE apply kora.

Q98. Explainable AI (XAI) ki? Model-er decision-ke manusher bujhar moto explain kora — jemon SHAP, LIME.

Q99. SHAP values ki? Game theory-based approach jeta proti feature koto contribute korlo ekta specific prediction-e — eta measure kore.

Q100. Production-e ML model deploy korar shomoy ki ki consider korba? Latency, scalability, model versioning, monitoring (drift detect), retraining pipeline.

Q101. Feature Store ki? Centralized system jekhane engineered feature store thake, reuse kora jay multiple model/team-e.

Q102. MLOps ki? ML model-er full lifecycle (development, deployment, monitoring, retraining) manage korar practice, DevOps-er ML version.

Q103. Ekta model production-e deploy korar por accuracy hothat kome gele first ki check korba? Data drift — input data distribution change hoise kina check kora.

14 Chapter 14: Cheat Sheet

14.1 14.1 One-Page ML Workflow

Collect → Clean → EDA → Feature Engineer → Select → Split
 → Scale → Train → Evaluate → Tune → Deploy

14.2 14.2 All Important Formulas

Concept	Formula
Linear Regression	$y = \beta_0 + \beta_1 x_1 + \dots + \epsilon$
MSE (cost)	$\frac{1}{n} \sum (y_i - \hat{y}_i)^2$
Ridge Penalty	$\lambda \sum \beta_j^2$
Lasso Penalty	$\lambda \sum \beta_j $
Sigmoid	$\frac{1}{1+e^{-z}}$
Binary Cross Entropy	$-\frac{1}{n} \sum [y \log \hat{y} + (1-y) \log(1-\hat{y})]$
Entropy	$-\sum p_i \log_2 p_i$
Gini Index	$1 - \sum p_i^2$
Information Gain	$Entropy_{parent} - \sum \frac{n_c}{n_p} Entropy_{child}$
Euclidean Distance	$\sqrt{\sum (p_i - q_i)^2}$
MAE	$\frac{1}{n} \sum y_i - \hat{y}_i $
RMSE	$\sqrt{MSE}$
R ²	$1 - \frac{\sum (y_i - \hat{y}_i)^2}{\sum (y_i - \bar{y})^2}$
Accuracy	$\frac{TP+TN}{TP+TN+FP+FN}$
Precision	$\frac{TP}{TP+FP}$
Recall	$\frac{TP}{TP+FN}$
F1 Score	$2 \times \frac{P \times R}{P+R}$
Standardization	$\frac{x-\mu}{\sigma}$
Normalization	$\frac{x-x_{min}}{x_{max}-x_{min}}$

14.3 14.3 Algorithm Comparison (Regression)

Algorithm	Handles Non-linearity	Regularized	Interpretable
Linear Regression	No	No	High
Polynomial Regression	Yes	No	Medium
Ridge	No	Yes(L2)	High
Lasso	No	Yes(L1)	High
ElasticNet	No	Yes(L1+L2)	High
Decision Tree	Yes	No	High
Random Forest	Yes	Ensemble	Medium

14.4 14.4 Algorithm Comparison (Classification)

Algorithm	Speed	Scaling Needed	Interpretable
Logistic Regression	Fast	Yes	High
Decision Tree	Fast	No	High
Random Forest	Medium	No	Medium
KNN	Slow(predict)	Yes	Medium
Naive Bayes	Very Fast	No	Medium
SVM	Slow(train)	Yes	Low

14.5 14.5 Metric Comparison

Task	Metric Use Korba
Regression, outlier-sensitive na chao	MAE
Regression, boro error penalize korte chao	RMSE
Regression, variance explained bujhte chao	R ² / Adjusted R ²
Balanced Classification	Accuracy
False Positive costly	Precision
False Negative costly	Recall
Imbalanced Classification	F1 Score, AUC-ROC

14.6 14.6 Bagging vs Boosting

Bagging (parallel): Tree1 Tree2 Tree3 → Average/Vote

Boosting (sequential): Tree1 → error → Tree2 → error → Tree3 → Final

14.7 14.7 Flowchart: Which Algorithm to Choose?

Target continuous? —Yes—> Regression

| No



Target category? —Yes—> Classification

|



No target/label? —Yes—> Unsupervised (Clustering)

Regression choose:

Simple relationship? → Linear Regression

Many correlated features? → Ridge/ElasticNet

Need feature selection? → Lasso

Non-linear? → Decision Tree / Random Forest / XGBoost

Classification choose:

Need probability + interpretable? → Logistic Regression

Non-linear + interpretable? → Decision Tree

High accuracy, less interpretability ok? → Random Forest / XGBoost

Small dataset, text data? → Naive Bayes

High-dimensional, complex boundary? → SVM

14.8 14.8 Common Interview Tricks

- **“Ridge vs Lasso” asle** — always mention: Ridge kokhono coefficient 0 kore na, Lasso pare (feature selection).

- **“Bias-Variance” asle** — always draw the tradeoff curve concept: High Bias=Underfit, High Variance=Overfit.
- **“Precision vs Recall” asle** — always link real example: Spam(Precision important), Disease(Recall important).
- **“Bagging vs Boosting” asle** — always mention: parallel vs sequential, variance↓ vs bias↓.
- **“Overfitting rodh korar way” jiggesh korle** — bolo: Regularization, Cross-validation, More data, Simpler model, Early Stopping, Pruning, Dropout(neural net).
- **“Confusion Matrix theke shob metric ber koro” bolle** — always TP/FP/TN/FN diye shuru koro, tarpor formula apply koro.

14.9 Final Quick Revision (Whole Book)

AI $\supset$ ML $\supset$ DL

Supervised = Regression + Classification

Regression metric: MAE, MSE, RMSE, R^2 , Adj R^2

Classification metric: Accuracy, Precision, Recall, F1, AUC

Ensemble: Bagging(Random Forest) | Boosting(AdaBoost, GBM, XGBoost, LightGBM, CatBoost)

Tuning: GridSearch, RandomSearch, Bayesian, Early Stopping

Validation: Hold-Out, K-Fold, Stratified K-Fold, LOOCV

Feature Engineering: Missing→impute, Outlier→IQR, Scale→Standard/MinMax, Encode→Label/OneHot/Target

Congratulations! Tumi ekhon puro Machine Learning fundamentals cover korse — beginner theke interview-ready porjonto. Practice koro, Kaggle e project banao, ar interview question gula bar bar revise koro. Best of luck!